

suite_id	representative_suite_id	key_handshake_id	key_handshake	suite_data_aead_id	suite_data_aead	handshake_runtime_aead	key
NIST Level L1							
khs-classicmceliece348864-falcon512	cs-classicmceliece348864-aesgcm-falcon512	khs-classicmceliece348864-falcon512	Classic-McEliece-348864+Falcon-512	dap-aesgcm	aesgcm	none (handshake-only)	Class
khs-classicmceliece348864-mlds44	cs-classicmceliece348864-aesgcm-mlds44	khs-classicmceliece348864-mlds44	Classic-McEliece-348864+ML-DSA-44	dap-aesgcm	aesgcm	none (handshake-only)	Class
khs-classicmceliece348864-sphincs128s	cs-classicmceliece348864-aesgcm-sphincs128s	khs-classicmceliece348864-sphincs128s	Classic-McEliece-348864+SPHINCS+-SHA2-128s-simple	dap-aesgcm	aesgcm	none (handshake-only)	Class
khs-hqc128-falcon512	cs-hqc128-aesgcm-falcon512	khs-hqc128-falcon512	HQC-128+Falcon-512	dap-aesgcm	aesgcm	none (handshake-only)	HQC
khs-hqc128-mlds44	cs-hqc128-aesgcm-mlds44	khs-hqc128-mlds44	HQC-128+ML-DSA-44	dap-aesgcm	aesgcm	none (handshake-only)	HQC
khs-hqc128-sphincs128s	cs-hqc128-aesgcm-sphincs128s	khs-hqc128-sphincs128s	HQC-128+SPHINCS+-SHA2-128s-simple	dap-aesgcm	aesgcm	none (handshake-only)	HQC
khs-mlkem512-falcon512	cs-mlkem512-aesgcm-falcon512	khs-mlkem512-falcon512	ML-KEM-512+Falcon-512	dap-aesgcm	aesgcm	none (handshake-only)	ML-K
khs-mlkem512-mlds44	cs-mlkem512-aesgcm-mlds44	khs-mlkem512-mlds44	ML-KEM-512+ML-DSA-44	dap-aesgcm	aesgcm	none (handshake-only)	ML-K
khs-mlkem512-sphincs128s	cs-mlkem512-aesgcm-sphincs128s	khs-mlkem512-sphincs128s	ML-KEM-512+SPHINCS+-SHA2-128s-simple	dap-aesgcm	aesgcm	none (handshake-only)	ML-K
NIST Level L3							
khs-classicmceliece460896-mlds65	cs-classicmceliece460896-aesgcm-mlds65	khs-classicmceliece460896-mlds65	Classic-McEliece-460896+ML-DSA-65	dap-aesgcm	aesgcm	none (handshake-only)	Class
khs-classicmceliece460896-sphincs192s	cs-classicmceliece460896-aesgcm-sphincs192s	khs-classicmceliece460896-sphincs192s	Classic-McEliece-460896+SPHINCS+-SHA2-192s-simple	dap-aesgcm	aesgcm	none (handshake-only)	Class
khs-hqc192-mlds65	cs-hqc192-aesgcm-mlds65	khs-hqc192-mlds65	HQC-192+ML-DSA-65	dap-aesgcm	aesgcm	none (handshake-only)	HQC
khs-hqc192-sphincs192s	cs-hqc192-aesgcm-sphincs192s	khs-hqc192-sphincs192s	HQC-192+SPHINCS+-SHA2-192s-simple	dap-aesgcm	aesgcm	none (handshake-only)	HQC
khs-mlkem768-mlds65	cs-mlkem768-aesgcm-mlds65	khs-mlkem768-mlds65	ML-KEM-768+ML-DSA-65	dap-aesgcm	aesgcm	none (handshake-only)	ML-K
khs-mlkem768-sphincs192s	cs-mlkem768-aesgcm-sphincs192s	khs-mlkem768-sphincs192s	ML-KEM-768+SPHINCS+-SHA2-192s-simple	dap-aesgcm	aesgcm	none (handshake-only)	ML-K
NIST Level L5							
khs-classicmceliece8192128-falcon1024	cs-classicmceliece8192128-aesgcm-falcon1024	khs-classicmceliece8192128-falcon1024	Classic-McEliece-8192128+Falcon-1024	dap-aesgcm	aesgcm	none (handshake-only)	Class
khs-classicmceliece8192128-mlds87	cs-classicmceliece8192128-aesgcm-mlds87	khs-classicmceliece8192128-mlds87	Classic-McEliece-8192128+ML-DSA-87	dap-aesgcm	aesgcm	none (handshake-only)	Class
khs-classicmceliece8192128-sphincs256s	cs-classicmceliece8192128-aesgcm-sphincs256s	khs-classicmceliece8192128-sphincs256s	Classic-McEliece-8192128+SPHINCS+-SHA2-256s-simple	dap-aesgcm	aesgcm	none (handshake-only)	Class
khs-hqc256-falcon1024	cs-hqc256-aesgcm-falcon1024	khs-hqc256-falcon1024	HQC-256+Falcon-1024	dap-aesgcm	aesgcm	none (handshake-only)	HQC
khs-hqc256-mlds87	cs-hqc256-aesgcm-mlds87	khs-hqc256-mlds87	HQC-256+ML-DSA-87	dap-aesgcm	aesgcm	none (handshake-only)	HQC
khs-hqc256-sphincs256s	cs-hqc256-aesgcm-sphincs256s	khs-hqc256-sphincs256s	HQC-256+SPHINCS+-SHA2-256s-simple	dap-aesgcm	aesgcm	none (handshake-only)	HQC
khs-mlkem1024-falcon1024	cs-mlkem1024-aesgcm-falcon1024	khs-mlkem1024-falcon1024	ML-KEM-1024+Falcon-1024	dap-aesgcm	aesgcm	none (handshake-only)	ML-K
khs-mlkem1024-mlds87	cs-mlkem1024-aesgcm-mlds87	khs-mlkem1024-mlds87	ML-KEM-1024+ML-DSA-87	dap-aesgcm	aesgcm	none (handshake-only)	ML-K
khs-mlkem1024-sphincs256s	cs-mlkem1024-aesgcm-sphincs256s	khs-mlkem1024-sphincs256s	ML-KEM-1024+SPHINCS+-SHA2-256s-simple	dap-aesgcm	aesgcm	none (handshake-only)	ML-K

Table 2: AEAD Benchmark Results (All 9 AEADs, Encrypt/Decrypt, Grouped by NIST Level)

aead_name	key_tier	operation	detector	payload_size	iterations	duration_s	mean_us	median_us	throughput_ops	baseline_power_w	avg_gross_power_w	avg_net_power_w	total_net_energy_j	per_op_energy_uj	power_samples	sample_rate_hz
NIST Level L1																
AES-128-CCM	L1	decrypt	Baseline	256	617498	15.0	18.534	18.482	41165.6	3.0939	4.0671	0.9732	13.020618	21.086	426	28.0
AES-128-CCM	L1	encrypt	Baseline	256	613449	15.0	18.626	18.537	40896.1	3.0939	4.0613	0.9675	13.393722	21.833	441	29.0
AES-128-GCM	L1	decrypt	Baseline	256	979358	15.0	9.509	9.5	65288.2	3.091	4.0895	0.9985	15.225962	15.547	844	56.0
AES-128-GCM	L1	encrypt	Baseline	256	1005929	15.0	9.281	9.222	67060.8	3.091	4.2896	1.1985	16.892357	16.793	1008	67.0
Ascon-128a	L1	decrypt	Baseline	256	5282	15.0	2832.443	2827.182	352.1	3.1023	3.772	0.6697	10.346022	1958.732	119	8.0
Ascon-128a	L1	encrypt	Baseline	256	5089	15.0	2939.758	2927.033	339.2	3.1023	3.7788	0.6765	10.012135	1967.407	119	8.0
NIST Level L3																
AES-192-CCM	L3	decrypt	Baseline	256	566226	15.0	20.691	20.629	37746.8	3.0964	4.0131	0.9167	13.292661	23.476	306	20.0
AES-192-CCM	L3	encrypt	Baseline	256	559702	15.0	20.95	20.852	37313.1	3.0964	3.9119	0.8155	12.107978	21.633	387	26.0
AES-192-GCM	L3	decrypt	Baseline	256	954064	15.0	9.972	9.963	63602.2	3.1047	4.1245	1.0198	15.213961	15.946	826	55.0
AES-192-GCM	L3	encrypt	Baseline	256	975582	15.0	9.703	9.648	65038.0	3.1047	4.2662	1.1615	16.10444	16.508	1034	69.0
NIST Level L5																
AES-256-CCM	L5	decrypt	Baseline	256	520498	15.0	23.039	22.963	34699.1	3.1418	4.004	0.8622	12.232011	23.501	290	19.0
AES-256-CCM	L5	encrypt	Baseline	256	518279	15.0	23.24	23.148	34551.5	3.1418	4.0044	0.8626	11.898871	22.958	377	25.0
AES-256-GCM	L5	decrypt	Baseline	256	924541	15.0	10.413	10.389	61633.5	3.1032	4.1378	1.0346	15.334236	16.586	750	50.0
AES-256-GCM	L5	encrypt	Baseline	256	954059	15.0	10.113	10.092	63603.0	3.1032	4.2282	1.125	16.066401	16.84	1390	93.0
AES-GCM-256(tunnel)	L5	decrypt	Baseline	256	924148	15.0	10.406	10.389	61608.1	3.0973	4.2928	1.1955	16.592771	17.955	648	43.0
AES-GCM-256(tunnel)	L5	encrypt	Baseline	256	952838	15.0	10.107	10.056	63521.8	3.0973	4.282	1.1847	16.723364	17.551	631	42.0
ChaCha20-Poly1305	L5	decrypt	Baseline	256	1165601	15.0	7.044	7.037	77703.7	3.1087	4.2057	1.0971	15.727596	13.493	682	45.0
ChaCha20-Poly1305	L5	encrypt	Baseline	256	1164734	15.0	6.84	6.815	77647.9	3.1087	4.179	1.0703	15.289411	13.127	1152	77.0